

U6 Theme Reading —The Blood and Blood Circulation

Medical English | 心血管系统 II: 血液与血液循环

最后修订: 2026 年 6 月 30 日

考点速览

- 世界献血者日 (WBDD): June 14, 2005 年起, WHO + Red Cross
- 血液物理特性 (5 项) 与全血组成 (Plasma 55% + Formed Elements 45%)
- 红细胞 (biconcave discs)、白细胞 (5 种分类)、血小板结构与功能
- 血液三大功能: Transportation, Regulation, Protection
- 肺循环 (Pulmonary) vs. 体循环 (Systemic) + 血液循环 6 步路径
- 脉搏 (Pulse) 与血压 (Blood Pressure) 定义、血压测量步骤
- 静脉穿刺 (Venipuncture) 流程与化验报告 (Hematology/Biochemistry) 解读

血液物理特性与组成

物理特性: Metallic taste (金属味), Scarlet to dull red (鲜红 → 暗红, 含氧量决定), 5× thicker viscosity (比水粘稠 5 倍), Alkaline pH, Slightly higher temperature (略高于体温)

全血组成:

成分	占比/细分
Plasma (血浆) — 55%	Water 90% + Dissolved substances 10% (nutrients, salts, gases, hormones, plasma proteins)
Formed Elements — 45%	Erythrocytes/RBCs + Leukocytes/WBCs + Platelets/Thrombocytes

血细胞分类

红细胞 (**Erythrocytes**): biconcave discs (双凹圆盘状), 无细胞核, ferry oxygen to all cells

白细胞 (**Leukocytes**) 分类:

- **Granulocytes** (粒细胞): neutrophils (中性粒细胞), eosinophils (嗜酸性粒细胞), basophils (嗜碱性粒细胞)
- **Agranulocytes** (无粒细胞): monocytes (单核细胞), lymphocytes (淋巴细胞)

功能: defend the body against bacteria, viruses, parasites and tumor cells

血小板 (**Platelets/Thrombocytes**): irregularly shaped (形状不规则), 来源于 megakaryocyte (巨核细胞)

血液三大功能: **Transportation** (运输 O₂、营养物质、激素、废物), **Regulation** (调节体温、pH、体液平衡), **Protection** (免疫防御、凝血)

血液循环 (Blood Circulation)

循环	英文	特点
肺循环	Pulmonary circulation	low resistance, high capacitance bed (低阻力、高容量床)
体循环	Systemic circulation	relatively high resistance vascular bed (相对高阻力血管床)

循环路径 (6 步):

1. Oxygen-poor blood from body → **right atrium**

2. Right atrium → **right ventricle**
3. Right ventricle → **lungs** (release waste gases, pick up oxygen)
4. Oxygen-rich blood → **left atrium**
5. Left atrium → **left ventricle**
6. Left ventricle → all parts of the body

脉搏、血压与测量

Pulse (脉搏): 左心室每次收缩后, 弹性动脉交替扩张和回缩所产生的行进的压力波

Blood Pressure (血压): 左心室收缩期在动脉中产生的压力 + 心室舒张时动脉中残留的压力

血压测量步骤:

1. Wrap the **cuff** around patient's upper arm
2. Place **stethoscope** over brachial artery
3. Close valve on **bulb**, pump up to ≥ 150 mmHg, slowly deflate
4. First sound = **systolic** / Sound disappears = **diastolic**

静脉穿刺与化验报告

Venipuncture: Phlebotomist (抽血技师), antecubital fossa (肘前窝), tourniquet (止血带), make a fist (握拳), dressing/gauze (敷料/纱布) 按压防 hematoma/bruise (血肿/淤青)

化验报告关键指标:

缩写	全称/中文
FBC	Full Blood Count (全血计数)
Hb	Haemoglobin (血红蛋白)
HCT	Haematocrit (红细胞压积)
MCV	Mean Cell Volume (平均红细胞体积)
WCC	White Cell Count (白细胞计数)
Urea/Creatinine	尿素/肌酐
Na/K	钠/钾
Bilirubin/ALT	胆红素/谷丙转氨酶

结果描述: 偏高 = up, high, raised, elevated (↑) / 正常 = normal, unremarkable, within normal limits / 偏低 = down, low, reduced (↓)

重点词汇

English	Chinese
World Blood Donor Day / transfusion	世界献血者日 / 输血
whole blood / plasma	全血 / 血浆
formed elements	有形成分
erythrocyte (RBC) / leukocyte (WBC)	红细胞 / 白细胞
platelet / thrombocyte	血小板
granulocyte / agranulocyte	粒细胞 / 无粒细胞
neutrophil / eosinophil / basophil	中性粒细胞/嗜酸性粒细胞/嗜碱性粒细胞
monocyte / lymphocyte	单核细胞 / 淋巴细胞
albumin / clotting factors / antibody	白蛋白/凝血因子/抗体
pulmonary / systemic circulation	肺循环 / 体循环
systolic / diastolic pressure	收缩压 / 舒张压
venipuncture / phlebotomist	静脉穿刺 / 抽血技师

常见考题

血液物理特性填空、成分比例、白细胞种类归属 (granulocyte vs agranulocyte)、血液循环 6 步排序、化验指标缩写与全称配对、pulse/blood pressure/venipuncture 名词解释

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